

Advanced (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $x(x + 5) + 6$
- (C) $(x + 1)(x + 4)$
- (D) $x^2 + 5x$
- (E) $x^2 + 6$

Q2. Factor $x^2 - 7x + 12$.

- (A) $x(x - 3) - 4$
- (B) $x(x - 4) - 3$
- (C) $x(x - 7) + 12$
- (D) $x(x - 3) + 4$
- (E) $(x - 3)(x - 4)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $x(x + 5) - 3$
- (B) $x(x + 3) - 5$
- (C) $x(x - 5) + 3$
- (D) $x(x + 3) + 5$
- (E) $(x + 5)(x - 3)$

Q4. Factor the trinomial $x^2 - 9x + 20$.

- (A) $x(x - 4) - 5$
- (B) $x(x - 5) - 4$
- (C) $x(x - 4) + 5$
- (D) $x(x - 5) + 4$
- (E) $(x - 4)(x - 5)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $x(x + 4) + 6$
- (B) $x(x + 6) + 4$
- (C) $x(x + 4) - 6$
- (D) $x(x + 6) - 4$
- (E) $(x + 4)(x + 6)$

Q6. Factor $x^2 - x - 12$.

- (A) $x(x - 4) + 3$
- (B) $x(x + 3) - 4$
- (C) $x(x + 4) - 3$
- (D) $x(x - 3) - 4$
- (E) $(x - 4)(x + 3)$

Q7. Factor the trinomial $x^2 + 8x + 16$.

- (A) $x(x + 4) + 4$
- (B) $x(x + 8) + 4$
- (C) $x(x + 4) - 4$
- (D) $x(x + 8) - 4$
- (E) $(x + 4)^2$

Q8. Factor the trinomial $x^2 - 6x + 8$.

- (A) $x(x - 2) - 4$
- (B) $x(x - 4) - 2$
- (C) $x(x - 8) + 2$
- (D) $x(x - 2) + 4$
- (E) $(x - 4)(x - 2)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $2x^2 + 5x - 3$. Show all steps and justify your answer.

Q10. Factor the trinomial $3x^2 - 2x - 1$. Show all steps and justify your answer.

Chef's Tips

- Check for the correct factoring of trinomials
- Verify the application of different factoring methods
- Ensure students provide clear justifications for their answers